

1. Find the page fault ratio and hit ratio using FIFO, LRU, MRU and optimal algorithm with memory size = 3 and sequence is 3 2 1 3 4 1 6 2 4 3 4 2 1 4 5 2 1 3 4

2 1 3 4

n:- Number of frames = 3

Total page references = 19

$$\text{Hit ratio} = \frac{\text{Hit}}{19}$$

$$\text{Fault ratio} = \frac{\text{Fault}}{19}$$

1. FIFO

3 2 1 3 4

	3	2	1	3	4	1	6	2	4	3	4	2	1	4	5	2	1	3	4
1	-	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	-	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	4	4	6	6	4	3	4	4	4	4	5	5	5	3	4
	x	x	x	✓	x	✓	x	✓	x	x	x	✓	✓	✓	x	✓	✓	x	x

Hit = 8

Faults = 11

$$\text{Hit ratio} = \frac{8}{19}$$

$$\text{Fault ratio} = \frac{11}{19}$$

2. LRU

	3	2	1	3	4	1	6	2	4	3	4	2	1	4	5	2	1	3	4
1	-	-	1	1	1	1	1	4	4	4	4	4	4	4	4	4	1	1	1
2	-	2	2	2	4	4	2	2	2	2	2	2	2	2	5	5	5	3	3
3	3	3	3	3	3	6	6	6	3	3	3	3	3	1	1	1	2	2	4
	x	x	x	✓	x	✓	x	x	x	x	x	✓	✓	x	✓	x	x	x	x

Hit = 5

Faults = 14

$$\text{Hit ratio} = \frac{5}{19}$$

$$\text{Hit ratio} = \frac{5}{19}$$

3. MRU

	3	2	1	3	4	1	6	2	4	3	4	2	1	4	5	2	1	3	4
1	-	-	1	1	1	1	6	6	6	6	6	6	6	6	6	6	6	6	6
2	-	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	4	4	4	4	4	4	4	4	4	4	4	5	2	2	2
	x	x	x	✓	x	✓	x	✓	✓	x	x	✓	x	✓	x	x	✓	x	x

Hits = 7

page faults = 12

Hit ratio = $7/12$

Page Fault Ratio = $12/19$

4. optimal page Replacement:-

	3	2	1	3	4	1	6	2	4	3	4	2	1	4	5	2	1	3	4
1	-	-	1	1	1	1	6	6	6	3	3	3	1	1	1	1	1	1	1
2	-	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	4	4	4	4	4	4	4	4	4	4	5	5	5	3	4
	x	x	x	✓	x	✓	✓	✓	✓	x	✓	✓	x	✓	x	✓	✓	x	x

Hits = 9

Faults = 10

Hit Ratio = $9/19$

Fault Ratio = $10/19$

Efficient → optimal

APPLY FIFO, LRU and MRU and check whether Belady's anomaly exists or not for physical memory 3 frames and 4 frames

1 2 3 4 1 2 5 1 2 3 4 5

A:- Total page references = 12
No. of frames = 3

	1	2	3	4	1	2	5	1	2	3	4	5
F_3	-	-	3	3	3	3	3	3	3	3	3	3
F_2	-	2	2	2	2	2	2	2	2	2	2	2
F_1	1	1	1	4	1	1	5	1	1	1	4	5
	x	x	x	x	x	✓	x	x	✓	✓	x	x

Faults = 9

Hits = 3

Hit ratio = $3/12$

Faults Ratio = $9/12$

For 4 frames:-

	1	2	3	4	1	2	5	1	2	3	4	5
F_4	-	-	-	4	4	4	4	4	4	3	3	3
F_3	-	-	3	3	3	3	3	3	3	2	2	2
F_2	-	2	2	2	2	2	2	2	2	1	1	5
F_1	1	1	1	1	1	1	5	5	5	5	4	4
	x	x	x	x	✓	✓	x	x	x	x	x	x

Faults = 10

Hits = 2

Hit Ratio = $2/12$

Fault ratio = $10/12$

Belady's Anomaly exists for FIFO because
3 frames → 9 faults
4 frames → 10 faults.

So, increasing frames increases faults.

LRU For 3 frames

	1	2	3	4	1	2	5	1	2	3	4	5
F ₃	-	-	3	3	3	2	2	2	2	2	2	5
F ₂	-	2	2	2	1	1	1	1	1	1	4	4
F ₁	1	1	1	4	4	4	5	5	5	3	3	2
	x	x	x	x	x	x	x	✓	✓	x	x	x

Faults = 10

Hits = 2

Hit Ratio = 2/12

Fault Ratio = 10/12

MRU

	1	2	3	4	1	2	5	1	2	3	4	5
F ₃	-	-	3	4	4	4	4	4	4	4	4	4
F ₂	-	2	2	2	2	2	5	5	5	5	5	5
F ₁	1	1	1	1	1	1	1	1	2	3	3	3
	x	x	x	x	✓	✓	x	✓	x	x	✓	✓

Faults = 7

Hits = 3

Fault ratio = 7/12

Hit Ratio = 3/12

most efficient → MRU